

Kinds as Projectibility Profiles

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Abstract

Anti-essentialist theories of kinds explain how categories can be real without sharing an essence. Homeostatic property cluster theory gives the central Boydian answer, but HOMEOSTATIC now covers several different achievements: stable co-occurrence, network order, maintenance, and corrective control. This paper proposes a more portable vocabulary. Kinds are projectibility profiles: structured patterns of properties, diagnostics, stabilizers, limits, and failure modes that license specified inferences under specified conditions. Stabilizing mechanisms matter because they explain why those projections work. Homeostatic control is a demanding subcase, not the default form of kindhood.

Keywords: natural kinds; projectibility; homeostatic property clusters; classification; philosophy of science

I INTRODUCTION

Anti-essentialist kind theory begins from a familiar pressure. Many categories that matter to science, grammar, social inquiry, and ordinary reasoning don't share an essence. Their members needn't have one property in common, and their boundaries may be graded, historically contingent, or context-sensitive. Still, some of these categories support reliable inference. They let us predict, explain, intervene, measure, compare, or learn.

Boyd's homeostatic property cluster framework made that combination possible. A kind can be real when a cluster of properties supports induction and explanation because classificatory practice has been accommodated to worldly structure rather than because every member shares a defining essence (Boyd, 1991, 1999). That insight remains the starting point. But the word HOMEOSTATIC has been asked to do too much. In different contexts it can name a stable profile, a network of relations, a maintaining process, or a corrective-control system.

Those are not the same achievement. A profile can be stable without a known maintainer. It can be ordered without being maintained. It can be maintained without being controlled. And it can support useful projection without earning the stricter label HOMEOSTATIC. Slater's stable property cluster account presses the stability point (Slater, 2015); Khalidi's causal-network account presses

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the order point (Khalidi, 2013, 2018). The lesson is not that Boyd was wrong, but that the evidential work needs sharper labels.

This paper proposes one such label: **PROJECTIBILITY PROFILE**. A kind, on this view, is a structured pattern of properties, diagnostics, stabilizers, limits, and failure modes that licenses specified inferences under specified conditions. A projectibility profile is not a merely conventional representation. Nor is it the worldly pattern in isolation from inquiry. It's an analytic specification of the worldly pattern in virtue of which a category supports projection.

The priority of projectibility is methodological and explanatory. We start by asking what the category lets us infer, then ask what profile supports that inference, what relations make the profile non-accidental, what stabilizes it, and where it fails. The order of inquiry doesn't imply that projectibility is metaphysically prior to every stabilizing structure. It says that a mechanism matters to kind theory because it explains projection, not because mechanism talk by itself makes a category real.

The paper is deliberately synthetic. Section 2 defines the profile package and distinguishes profiles, stabilizers, maintainers, and controllers. Section 3 explains why the projection question comes first. Section 4 shows how the vocabulary travels across explanatory levels without turning every case into a biological homeostasis case. Section 5 gives the vocabulary teeth by making demotion and failure modes explicit. Section 6 states the payoff.

2 THE PROFILE PACKAGE

A projectibility profile answers six questions. First, what's the projection target? Second, what observable profile supports it? Third, what relations make the profile non-accidental? Fourth, what stabilizes the profile across relevant contexts or timescales? Fifth, what failure modes mark the profile's limits? Sixth, under which scope conditions is the profile projectible?

The first question matters because recurrence is cheap. A list of co-occurring properties becomes theoretically interesting only when it licenses some further expectation. The expectation may be predictive, explanatory, classificatory, or intervention-guiding. It may concern a biological process, a grammatical pattern, a measurement practice, or a social role. But without a declared projection target, the category remains a description of observed association.

The second question asks what the profile is. A profile is the structured pattern whose partial observation licenses specified inferences. In a biological case, it may include morphology, development, ecology, and reproductive relations. In a grammatical case, it may include distributional diagnostics, semantic tendencies, diachronic sources, acquisition patterns, and mismatch residue. In a social case, it may include institutional marks, practical responses, uptake conditions, sanctions, and repair paths. The pattern needn't be one-level or one-mechanism.

The third question asks why the profile is non-accidental. Khalidi's causal network account supplies one important model: kinds are not mere concatenations of properties, but nodes in ordered networks where some properties help explain others (Khalidi, 2018). In other domains, the relevant order may be functional, conventional, diachronic, normative, algorithmic, or evidential rather than narrowly causal. The level has to be marked. Otherwise a social coordination pattern, a cognitive mechanism, and a corpus distribution begin to explain one another by equivocation.

The fourth question asks what stabilizes the profile. I use **STABILIZER** for any relation, constraint, mechanism, or practice that helps make a projection reliable. Stabilizers needn't be active processes. Some are structural dependencies, selectional constraints, shared standards, ecological regularities, or

measurement conventions. What matters is that they help explain why observing part of the profile gives reason to expect the rest.

A MAINTAINER is more specific. It's an active stabilizer that preserves or regenerates the profile over time or under perturbation. A maintainer may be developmental, social, institutional, biological, or interactional. It can keep a profile available without registering each departure as an error or returning a system to a fixed value.

A CONTROLLER is more specific again. It's a maintainer with perturbation-sensitive organization that preserves a higher-scale relation through lower-scale variation. On the stricter use defended in Reynolds (2026a), the term HOMEOSTATIC belongs here. Homeostasis is not a synonym for recurrence, order, or maintenance. It marks the demanding case where corrective organization preserves a relation despite perturbation.

The sixth question concerns scope. A profile is projectible under some range of conditions, contrasts, and purposes. This doesn't make kindhood arbitrary. Purposes affect which available projections matter for an inquiry; they don't create the availability of those projections. A profile may support one family of inferences and fail for another. That gives us a reason to localize the claim, not a reason to deny realism wherever interests enter.

3 WHY PROJECTIBILITY COMES FIRST

Goodman's new riddle made PROJECTIBILITY the name for a hard problem: some predicates that fit observed cases still don't license projection to unobserved ones (Goodman, 1955). The present use of PROJECTIBLE is Boydian rather than Goodmanian in one central respect. It doesn't treat projectibility as a matter of entrenchment in our predicate-using history. It asks what worldly order makes a category a reliable guide to further expectation.

That realist turn is the point of Boyd's anti-essentialism. We don't need a shared essence to explain why a category supports induction. We need a non-accidental profile that our classificatory practice has partly learned to track (Boyd, 1991). Projectibility is therefore the target of the kind claim. Mechanisms, networks, histories, standards, and practices matter because they explain why projection works.

This order blocks a common slide. Recurrence gets promoted to stability, stability to stabilization, stabilization to homeostasis, and homeostasis to kindhood. The profile vocabulary reverses that order. Start with the inference. Ask what profile licenses it. Then ask which stabilizing claims have actually been earned.

The point is not that mechanisms are secondary in the world. During inquiry, we may have good evidence that a profile is projectible before we know what maintains it. The reliable projection still depends on whatever worldly relations make the profile non-accidental. The epistemic order and the grounding order can come apart. The proposed discipline is to avoid treating the unknown ground as if it had already been identified, named, and promoted to homeostasis.

This also explains why stable property clusters are important but not enough. Slater is right that natural kindness needn't wait for a named homeostatic mechanism (Slater, 2015). But a stable profile still has to be stable for some projection. A category may show recurring co-instantiation and yet fail to support the inference for which it was introduced. Conversely, a category may support modest local projections while failing to support broader ones. Both results are informative.

Purpose-relativity enters here with care. A profile is projectible for a family of questions, contrasts,

and evidential practices. In physiology, a useful profile may license intervention and measurement. In grammar, it may license expectations about distributions, manipulations, and borderline cases. In social ontology, it may license expectations about uptake, status, sanctions, or institutional repair. The practice selects which projection matters. It doesn't make the projection true.

A projectibility profile therefore differs from a checklist. Checklists record features. Profiles connect features to expectations under scope conditions. A checklist asks whether an item has properties P_1 , P_2 , and P_3 . A profile asks whether observing P_1 and P_2 , in the relevant setting, licenses expectation of P_3 , and what would defeat that expectation.

The difference matters for explanatory economy. A mechanism-first paper can finish by naming structures that hold a cluster together. A profile-first paper has to say what those structures let us project, what failures would demote the claim, and whether the advertised stabilizer operates at the right explanatory level. The contribution is a discipline for using categories as guides to further expectation.

4 PROFILES ACROSS LEVELS

The profile vocabulary is meant to travel, but not by making every domain look biological. Profiles can be behavioral, algorithmic, corpus-distributional, model-internal, institutional, normative, social-practical, or biological. The level matters because evidence at one level rarely licenses projection at another without a bridge claim.

Linguistic categories make the point concrete. In the deitality analysis of English, the point of introducing the category isn't to rename every definite form. The category captures a structural profile that supports further expectations: which diagnostics should cluster, which semantic mismatches should leave distributional residue, which contrasts should shift under prosody or specificity manipulation, and which variation should appear across dialects and acquisition (Reynolds, 2025a). The category earns its place by those projections.

The same discipline applies to typological comparison. A comparative concept can remain an instrument for measurement. It becomes a naturalized kind claim only if the comparandum shows a profile that travels across languages, supports out-of-sample projection, and comes with demotion criteria when the profile proves too thin (Reynolds, 2025b). Here the stabilizing background may include acquisition, processing, diachrony, discourse pressure, and measurement design. None of those should be collapsed into a universal language-internal category.

Truth-tracking supplies a different portability test. A representation's success can depend on partially coupled routes such as perception, testimony, coherence, measurement, intervention, error-correction, and practical success (Reynolds, 2026b). A large language model can participate strongly in some routes and weakly in others. The profile predicts failure signatures: fluent coherence may travel where coherence is the missing support, while retrieval or tools help only when they strengthen the needed stabilizer.

In each case, the same vocabulary applies without imposing the same mechanism. The deital profile is grammatical and distributional. The typological profile is comparative and methodological. The truth-tracking profile is epistemic and representational. Calling all three PROFILES doesn't erase those differences. It lets us ask the same disciplined questions: what gets projected, what pattern supports it, what stabilizes it, and where it fails.

Fields, practices, and institutions should be handled in the same way. They can be scope condi-

tions, stabilizing backgrounds, or parts of a profile. They should not be promoted to kinds merely because they organize inquiry or action. A field may condition which projections matter without itself being the projection target. A practice may stabilize a category without being the kind. Those are bridge claims, and they need their own evidence.

5 FAILURE MODES AND DEMOTION

The vocabulary is useful only if it can say no. Without failure modes, PROJECTIBILITY PROFILE would become another loose label for any interesting cluster. A profile view needs demotion rules: ways to say that a category is too thin, too local, wrongly leveled, or overpromoted.

The first failure mode is a thin profile. A thin profile has diagnostics that identify familiar cases but don't license further expectations. The category may still be convenient, but the kind claim is weak. In grammatical work, a label that sorts textbook examples but predicts nothing about mismatch, manipulation, acquisition, or variation is thin. In typology, a comparative label that cannot generalize beyond the coding sample is thin.

The second failure mode is wrong-level projection. This happens when evidence at one level is used to project at another without a bridge. A corpus distribution doesn't by itself establish a model-internal mechanism. A social-practical regularity doesn't by itself establish a biological disposition. A grammatical category in one language doesn't by itself establish a cross-linguistic comparandum. The projection may be possible, but the bridge has to be argued.

The third failure mode is decorative mechanism. A paper names a mechanism, but the mechanism doesn't underwrite any new projection. The result is explanatory decoration: a causal or functional story attached to a category whose inferential payoff remains unchanged. The profile test is simple. If removing the mechanism talk leaves the same predictions, the mechanism has not earned its place in the kind claim.

The fourth failure mode is over-homeostatization. Stability, network order, and maintenance are often real achievements. They should not be relabeled as homeostasis unless the evidence shows corrective organization preserving a higher-scale relation through lower-scale variation. The risk is especially high outside biology, where the attraction of the homeostasis metaphor can hide weak evidence. A maintained convention, a stable distribution, or a reliable measurement practice may be projectible without being homeostatic.

The fifth failure mode is scope overreach. A profile can be real and useful under one set of conditions while failing elsewhere. The error is to export the projection without fresh tests. A category projectible for discourse analysis may not be projectible for syntax. A model behavior projectible under benchmark conditions may not transfer to tool-using settings. A social status projectible inside one institution may not travel across institutions.

Demotion belongs inside realism. If kind claims are answerable to the projections they support, then failed projection should change the claim. Sometimes the right response is rejection: the category doesn't support the advertised inference. Sometimes localization is right: the category is projectible only under narrower scope conditions. Sometimes releveled is right: the pattern is real, but at a different explanatory level.

Profile talk joins promotion and demotion. A category earns stronger status when its profile supports reliable projection, when its stabilizers are identified at the right level, and when its limits are known. It loses status when those conditions fail. The framework therefore gives anti-essentialist

kind theory an exit procedure, not just a route into realism.

6 CONCLUSION

The slogan is compact: a category is a profile, stabilized by mechanisms, projectible under specified conditions. Each clause matters. A category is a profile because no single essence need define it. Stabilization matters because recurrence needs explanation. Projectibility matters because kind claims earn their scientific or grammatical value by licensing further expectation.

Homeostasis remains important, but it no longer has to carry every achievement. Some profiles are stable. Some are network ordered. Some are maintained. Some are controlled. Only the last deserves the strict homeostatic label. Keeping those tiers apart preserves Boyd's anti-essentialist realism while incorporating Slater's stability-first lesson and Khalidi's network discipline.

The result is a portable vocabulary for asking better questions. What does this category let us infer? What pattern supports that inference? What stabilizes the pattern? What would defeat the projection? And when should the category be demoted, localized, or rejected? Those questions are the work a theory of kinds has to do.

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